A DISSERTATION REPORT ON

TRENDPULSE — SENTIMENT AND KEYWORD ANALYSIS IN REAL-TIME SOCIAL MEDIA TREND ANALYZER

SUBMITTED IN PARTIAL FULFILMENT FOR THE REQUIREMENTS

OF THE DEGREE

BACHELOR OF ENGINEERING

IN

COMPUTER ENGINEERING

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CERTIFICATE

This is to Certified that, This Project report entitled.

Name Of Project

Submitted By

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PROJECT REPORT APPROVAL FOR B.E.

This project report entitled "**Natural Language Processing for Sentiment and Keyword Analysis in Social Media**" by **"**Dev Prasad", "Rohan Wajekar", "Samruddhi Dere", "Devika Jagtap", "Omkar Talekar.**"** is approved for the degree of **"Bachelor of Computer Engineering".**

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DECLARATION

We hereby declare that the technical report entitled **"Natural Language Processing for Sentiment and Keyword Analysis in Social Media"** is prepared as part of the TrendPulse — Real-Time Social Media Trend Analyzer project.

The report explains the application of Natural Language Processing techniques for analysing text-based social media content, identifying frequently occurring keywords, and determining the sentiment expressed in textual data.

The material has been prepared for academic purposes using relevant technical documentation, research publications, and the project's intended functionality. Any references to implementation, evaluation, and system behaviour should be interpreted according to the actual code and test results available for the project.

We understand that all sources used in preparing the report must be acknowledged appropriately.

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Finally, we thank our classmates, friends, and everyone who contributed suggestions and support during the project. Their assistance helped us understand how Natural Language Processing can be applied to social media data and how textual information can be converted into meaningful analytical insights.

ABSTRACT

Social media platforms generate large amounts of textual information every day through posts, comments, captions, reviews, and discussions. This information contains opinions, emotions, interests, and emerging topics that can help users understand public reactions and changing trends. However, manually analysing large collections of social media text is time-consuming and difficult because the data is unstructured, continuously changing, and often contains informal language.

Natural Language Processing (NLP) provides computational techniques for processing and analysing human language. In the Trend Pulse — Real-Time Social Media Trend Analyzer project, NLP supports the examination of text-based social media content through keyword analysis and sentiment classification.

The NLP component focuses on two principal tasks. The first is keyword analysis, which identifies frequently occurring terms and helps reveal the subjects receiving attention in the collected dataset. The second is sentiment analysis, which estimates whether a textual entry expresses a positive, negative, or neutral sentiment. The project uses Python-based data processing, Pandas for handling tabular data, and the VADER sentiment analysis approach to calculate sentiment polarity scores. The resulting information can be displayed through a Stream lit dashboard using visualizations created with Plotly.

The general workflow includes loading textual data, selecting relevant text fields, performing necessary text preparation, analysing frequently occurring words, calculating sentiment scores, assigning sentiment categories, and presenting the results visually. Engagement information, when available in the dataset, can be examined alongside textual results to provide additional context for understanding social media trends.

This report explains the fundamentals of NLP, the problem addressed by the project, its objectives, existing approaches, proposed system architecture, keyword extraction concepts, VADER sentiment analysis, implementation methodology, testing strategy, limitations, and possible future enhancements. It also discusses challenges such as sarcasm, spelling variations, slang, multilingual content, context-dependent expressions, and the difficulty of interpreting sentiment from short social media posts.

The NLP component demonstrates how language-processing techniques can transform unstructured text into structured analytical information. The resulting insights can support trend exploration, opinion monitoring, and social media data analysis. The system is intended as an analytical aid rather than a definitive measure of public opinion.

Keywords: Natural Language Processing, Sentiment Analysis, VADER, Keyword Frequency, Social Media Analytics, Python, Pandas, Stream lit, Plotly, Text Mining.

Table of Contents

ABSTRACT **6**
Chapter 1. INTRODUCTION **8**
Chapter 2. LITERATURE SURVEY **11**
Chapter 3. PROBLEM STATEMENT, OBJECTIVES AND SCOPE **13**
Chapter 4. SYSTEM ANALYSIS, FRAMEWORK AND ALGORITHMS **15**
Chapter 5. SYSTEM DESIGN AND DATABASE DESIGN **17**
Chapter 6. SOFTWARE AND HARDWARE SPECIFICATIONS **21**
Chapter 7. PERFORMANCE EVALUATION PARAMETER AND RESULT .. **24**
Chapter 8. IMPLEMENTATION PLAN AND FUTURE SCOPE **28**
REFERENCES AND BIBLIOGRAPHY **30**

CHAPTER 1
INTRODUCTION

1.1 Background of Natural Language Processing

Natural Language Processing is a branch of Artificial Intelligence that enables computers to process, analyse, and interpret human language. Human language is complex because the same word or sentence can have different meanings depending on context, tone, intention, and surrounding words. Computers traditionally work with structured information such as numbers, tables, and predefined data fields. However, social media content is usually expressed as free-form text. A person may write a product review, post an opinion about a public event, express satisfaction with a service, or comment on a trending topic. Such information cannot be fully understood through numerical analysis alone.

NLP provides techniques for examining this textual information. Common NLP applications include machine translation, speech recognition, text summarization, question answering, chatbots, document classification, information extraction, and sentiment analysis.

For example, consider the following social media statements:

1. "The new application is excellent."
2. "The new application is disappointing."
3. "The new application was released yesterday."

The first statement expresses positive sentiment, the second expresses negative sentiment, and the third is primarily informational. An NLP system can estimate these differences and organize the statements into useful categories. In the TrendPulse project, NLP is applied to text-based social media data to identify frequently occurring keywords and estimate the sentiment expressed in individual text entries

1.2 NLP in Social Media Analytics

Social media analytics involves collecting, processing, and interpreting information generated through social networking platforms and online communities. Textual content can reveal what people discuss, what subjects receive attention, and how users react to products, services, events, technologies, and public issues.

However, the volume of content makes manual inspection difficult. An analyst may need to examine hundreds or thousands of posts. Reading each entry individually would require significant time and would make consistent analysis challenging. NLP helps automate some of these tasks.

Keyword analysis identifies recurring words or expressions in a dataset. Sentiment analysis estimates the emotional polarity of textual entries. When combined with engagement information such as likes, comments, shares, or views, these results can provide additional context for interpreting the popularity of a topic.

For example, a topic may receive a large number of interactions while containing predominantly positive comments. Another topic may receive similar attention but generate substantial negative reactions. These situations can be investigated separately using textual and engagement-based analysis. It is important to understand that keyword frequency and sentiment are different measurements. A frequently

occurring word does not automatically represent a popular trend, and a positive sentiment score does not directly measure the popularity of a topic

1.3 Need for NLP in Trend Pulse

Trend Pulse is a social media trend analysis project designed to present collected social media information through an interactive dashboard.

Numerical information such as likes, comments, shares, and views can help describe audience engagement. However, these numbers do not directly reveal the meaning of the text or the opinions expressed by users.

For example, two posts may receive the same number of likes but discuss different subjects and express different opinions. Numerical engagement alone cannot identify these differences.

The NLP component addresses this gap by analysing the textual fields associated with social media entries.

Its principal functions are:

- Processing textual information from the available dataset.
- Identifying frequently occurring keywords.
- Estimating positive, negative, and neutral sentiment.
- Producing sentiment scores that can be interpreted alongside the text.
- Preparing structured results for dashboard visualization.
- Supporting the exploration of relationships between textual content and engagement.

The NLP module therefore complements the data-processing and visualization components of Trend Pulse.

1.4 Role of NLP in the Overall Project

The NLP component is one part of a larger analytical pipeline. The overall system can be understood through four broad stages.

Stage 1: Data input

Social media records are provided through the available project datasets. The records may contain platform names, topics, textual content, and engagement values.

Stage 2: Text analysis

The NLP component examines the textual content. It identifies keywords and calculates sentiment scores using the selected analysis approach.

Stage 3: Data integration

The resulting sentiment and keyword information is associated with the corresponding records. Other available fields may be used for filtering or comparison.

Stage 4: Visualization

The processed information is displayed through the TrendPulse dashboard. Charts and tables help users inspect keyword patterns, sentiment distributions, and related analytical results.

The current project uses Python, Pandas, VADER-based sentiment analysis, Streamlit, and Plotly within this workflow.

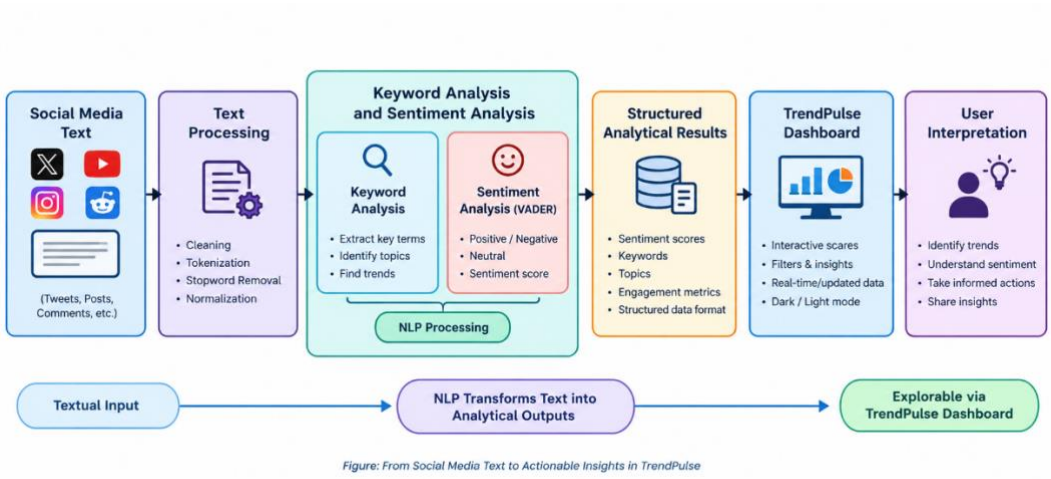


Figure 1.1: Role of NLP in Trend Pulse

CHAPTER 2
LITERATURE SURVEY AND EXISTING SYSTEM

2.1 Literature Survey

Real-time social media trend analysis is an intersection of Natural Language Processing (NLP), Sentiment Analysis, and Data Stream Mining. The goal is to extract public opinion, trending keywords, and sentiment polarity from high-velocity data like Twitter/X, Instagram, and Threads.

2. Literature Review.

Paper1: Sentiment Analysis on Twitter Data

Authors: Agarwal et al., 2011

Technique: They proposed a tree kernel-based model and feature-based SVM for Twitter sentiment classification

. **Findings:** POS-specific features and emoticons improve accuracy over bag-of-words. Achieved ~75% accuracy .

Relevance to TRENDPULSE: Base concept of sentiment classification for short, noisy social media text.

Limitation: Not real-time, did not handle trending topics.

Paper 2: Real-Time Sentiment Analysis of Twitter Streaming Data

Authors: Bifet & Frank, 2010

Technique: Used MOA (Massive Online Analysis) framework with incremental learning algorithms like Hoeffding Tree

Findings: Proved that batch learning fails on streaming data due to concept drift. Real-time models must be adaptive.

Relevance: Directly supports your real-time streaming architecture

Limitation: Low accuracy on sarcasm and slang.

Paper 3: Keyword Extraction for Trend Detection

Authors: Rose et al., 2010 (RAKE Algorithm)

Technique: RAKE - Rapid Automatic Keyword Extraction, a domain-independent algorithm.

Findings: Efficient keyword extraction without training data, based on word co-occurrence

Relevance: TRENDPULSE can use RAKE / YAKE / KeyBERT for trending keyword identification.

Limitation: RAKE fails to understand semantics.

Paper 4: BERT for Sentiment Analysis

Authors: Devlin et al., 2019 (Google BERT)

Technique: Transformer-based bidirectional pre-training

Findings: BERT outperformed all previous models on sentiment tasks with >93% accuracy on SST-2

Relevance: You can use Distil BERT or RoBERTa for more accurate sentiment analysis in TRENDPULSE instead of traditional VADER/Text Blob.

Limitation: High computational cost for real-time use, needs optimization.

Paper 5: Trend Detection in Social Media Streams

Authors: Mathioudakis & Koudas, 2010 – TwitterMonitor

Technique: Developed TwitterMonitor to detect emerging trends using burst detection in keywords.

Findings: Used queueing theory to detect sudden increase in keyword frequency.

Relevance: Core logic for your "Trending Pulse" detection module

Limitation: Only detects volume-based trends, not sentiment-based.

Paper 6: VADER - A Lexicon and Rule-Based Sentiment Tool

Authors: Hutto & Gilbert, 2014

Technique: VADER lexicon specifically attuned to social media - handles emojis, slang, CAPS, punctuation.

Findings: VADER performs well on social media data with F1 score of 0.96, better than humans in some cases.

Relevance: Best for lightweight, real-time sentiment analysis if you don't want to use heavy BERT models.

Limitation: Cannot handle context-dependent sarcasm

Paper 7: Real-time Event Detection using Social Media

Authors: Sakaki et al., 2010

Technique: Used Twitter as a social sensor for earthquake detection using SVM and Kalman filtering

Findings: Social media can detect events faster than traditional news

Relevance: Justifies why real-time social media analysis is important.

CHAPTER 3
PROBLEM STATEMENT, OBJECTIVES AND SCOPE

3.1 Problem Statement

Social media platforms generate large volumes of textual data containing information about users' interests, opinions, reactions, and discussions. Analysing this data manually is difficult because the text is unstructured, varies in length, and frequently contains informal expressions.

Traditional numerical analysis focuses mainly on measurable quantities such as views, likes, comments, and shares. Although these measurements help estimate engagement, they do not independently reveal whether the associated text expresses a positive, negative, or neutral opinion.

Similarly, reading every post individually to identify common topics is time-consuming and difficult to scale.

The problem addressed by the NLP component of TrendPulse is therefore to develop a practical text-analysis workflow that can process available social media text, identify frequently occurring keywords, estimate sentiment polarity, and present the resulting information in a form that supports further analysis.

The system uses rule-based and lexicon-based sentiment analysis rather than requiring a newly trained deep-learning model. This makes the selected approach suitable for a lightweight Python-based project, although it introduces limitations when interpreting complex language.

3.2 Objectives

1. Real-Time Data Collection

To collect and process live streaming data from social media platforms like Twitter/X, Instagram, and Threads using APIs.

2. Sentiment Analysis

To classify public opinion into Positive, Negative, and Neutral sentiments using NLP techniques (VADER / BERT) and handle emojis, slang, and short texts.

3. Keyword Extraction & Trend Detection

To automatically extract trending keywords, hashtags, and topics using algorithms like RAKE, YAKE, or Key BERT and detect sudden bursts in trends.

4. Live Visualization Dashboard

To create an interactive dashboard that shows:

Trending keywords (word cloud) Sentiment graph (positive vs negative) Trend volume over time Top hashtags

5. Early Event Detection

To identify emerging events or viral topics before they become mainstream news by analyzing frequency spikes.

6. Performance Optimization

To make the system lightweight and fast enough for real-time processing with low latency, even with high data velocity.

7. User-Friendly Reporting

To generate automated summary reports of trends and public sentiment for businesses, brands, or researchers were civic intelligence platform for organizing the reporting, analysis, prioritization, and monitoring of urban infrastructure issues. The system is designed to connect citizen reporting with administrative review through a common workflow. It aims to reduce dependence on unstructured complaint handling by maintaining organized incident records and supporting selected tasks through artificial intelligence and location-based analysis. The proposed system begins when a citizen submits a report containing details about a civic problem. The report may include a photograph, a textual description, and geographical coordinates. The information is then processed to identify the type of issue, examine whether a similar incident has already been reported, and estimate its priority. The resulting incident record can be made available to the relevant administrative interface for review, assignment, and progress monitoring.

The architecture divides the overall workflow into seven logical agents. Each agent is responsible for a specific operation, such as report ingestion, duplicate detection, severity assessment, resource allocation, status communication, repair verification, or escalation. These agents are logical software components and do not necessarily reqtlin and Jetpack Compose, with Google Cloud Firestore serving as the cloud data layer. AI-assisted image analysis and server-side processing are intended to support the intelligence functions. The exact operation of each component depends on its implementation and integration in the project. its initial submission to its eventual closure. Each stage produces information required by the next stage, allowing an incident to be processed in a structured manner. The process begins with citizen reporting. A resident identifies a civic problem and submits its description, photograph where available, and location information through the reporting interface. The submitted information forms the initial incident record.

The next stage involves issue classification and data organization. The system is designed to use AI-assisted image analysis and the submitted description to identify the likely category of the problem. The report is then compared with existing incidents using geographical proximity and other available information to identify possible duplicates. After classification and duplicate checking, the system evaluates the incident's priority. The proposed severity assessment considers relevant factors associated with the issue and produces information that can assist administrative decision-making. The incident can then be reviewed and assigned to the appropriate personnel or department according to the proposed allocation process. During the operational stage, the incident status is updated as work progresses. The citizen-facing interface is intended to display relevant status information, while the administrative interface supports monitoring of active complaints. Where repair verification is available, completion evidence can be reviewed before the incident is marked as resolved. The proposed escalation mechanism can identify incidents that remain unresolved beyond their applicable deadlines.

The final stage involves maintaining the incident record and its status history. This information supports traceability and can be used to review the complaint lifecycle. The workflow is intended to provide a consistent process, although actual notifications, automated assignment, verification, and escalation depend on the implemented component

CHAPTER 4
SYSTEM ANALYSIS, FRAMEWORK AND ALGORITHMS

4.1 Existing Approaches

Basic social media analysis often relies on numerical engagement measures, manual reading, or simple keyword searches.

Engagement metrics provide information about the amount of interaction associated with a post or topic. However, they do not independently reveal whether the discussion is positive, negative, or neutral.

Manual analysis can provide rich contextual understanding but requires time and human effort. Simple keyword searches can locate terms but may miss synonyms, variations, and expressions whose meaning depends on context.

4.2 Limitations of Traditional Methods

Traditional approaches may experience the following problems:

- Limited ability to process large volumes of text manually.
- Lack of consistent sentiment classification.
- Difficulty summarizing opinions across many comments.
- Inability to interpret engagement metrics in isolation.
- Dependence on predefined search terms.
- Difficulty identifying the broader vocabulary associated with a topic.

4.3 Proposed NLP Approach

The TrendPulse NLP component introduces automated text analysis into the social media analytics workflow.

The main steps are:

1. Obtain the text associated with a social media record.
2. Check whether the text is present and suitable for processing.
3. Apply appropriate text preparation where required.
4. Calculate sentiment using VADER.
5. Organize keyword information using the project's keyword-analysis process.
6. Store or expose the resulting information in a structured form.
7. Use the results for exploratory analysis and dashboard presentation.

This approach reduces the need for manual classification and supports consistent processing of records.

4.4 Advantages

The proposed approach offers several potential benefits.

Automation: The same analysis procedure can be applied to many records.

Consistency: Identical input text can be processed using the same rules and lexicon.

Speed: Lightweight rule-based analysis can be practical for short messages.

Interpretability: VADER exposes sentiment scores that can be inspected rather than providing only an opaque class label.

Integration: Text-derived features can be combined with likes, comments, shares, views, and topic information.

Extensibility: The design can later be enhanced with machine learning or contextual language models.

4.5 Difference Between Popularity and Sentiment

A high engagement score does not automatically mean a topic is positively received.

For example, a controversial product announcement may generate many comments and shares because people disagree with it.

Consequently, Trend Pulse should interpret engagement, sentiment, and keyword information as complementary measurements.

The combination can support exploratory analysis, but it should not be treated as proof of public opinion across an entire population.

CHAPTER 5
SYSTEM DESIGN AND DATABASE DESIGN

5.1 Overview

The NLP component operates as a processing layer between the social media text records and the analytical outputs used by TrendPulse. The broader application includes data files, data-processing logic, engagement analysis, and a dashboard. The NLP component concentrates on extracting textual insights. The exact software interfaces and file dependencies should be checked against the final source code before this report is submitted.

5.2 High-Level Architecture

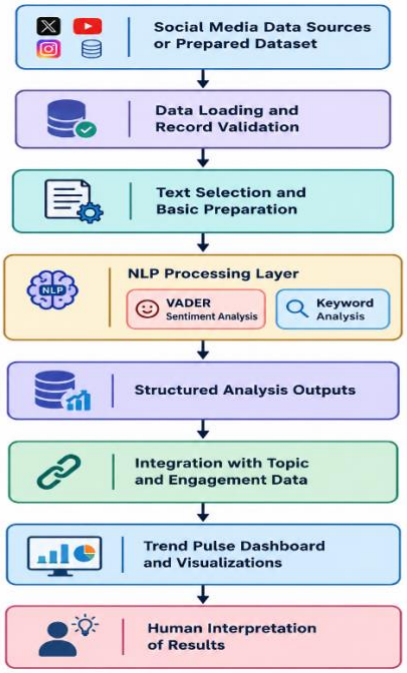


Figure 6.1: NLP Component Architecture

5.3 Data Input Layer

The input layer provides the records to be analyzed. The documented project structure includes fields such as:

- `platform`
- `topic`
- `text`
- `likes`
- `comments`
- `shares`
- `views`

The text field contains the language data relevant to NLP. The remaining fields provide metadata or engagement context. The application may also use prepared CSV files containing social media records and analysis outputs. The final implementation should identify the actual files used by the NLP pipeline.

5.4 Data Validation Layer

Before processing text, the system should verify that the required fields exist and that individual records are usable. Possible validation checks include:

- Missing text values.
- Empty strings.
- Unexpected data types.
- Duplicate records.
- Invalid numerical engagement values.
- Missing topic or platform information.

The presence of a field does not guarantee that its content is valid. For example, a text field may contain a null value or a string containing only whitespace. Validation helps prevent errors and makes the processing workflow more reliable.

5.5 Text Processing Layer

This layer prepares the text for the intended analytical task.Possible operations include handling missing values, normalizing whitespace, and identifying text that contains no meaningful content. More aggressive transformations, such as lowercasing, removing punctuation, or removing stop words, should be applied only when suitable for the task.

For sentiment analysis, excessive cleaning can remove useful information such as negation, punctuation emphasis, emoticons, and capitalization patterns. For keyword analysis, a different preprocessing configuration may be appropriate.

5.6 Sentiment Analysis Layer

The sentiment layer applies VADER to the selected text and obtains sentiment scores. VADER provides four commonly used values:

- Positive sentiment score.
- Negative sentiment score.
- Neutral sentiment score.
- Compound sentiment score.

The compound score is a normalized value between −1 and +1. It summarizes the overall sentiment orientation according to the VADER method. A sentiment label may then be assigned using documented thresholds. The selected thresholds must remain consistent throughout the application and evaluation.

5.7 Keyword Analysis Layer

The keyword layer identifies terms that may help summarize the content of the dataset. The actual method may involve frequency counting, filtering, aggregation, or another extraction procedure, depending on the current implementation. The project contains keyword-related data files. Their exact schema and generation procedure should be verified in the source code. A keyword list can help users identify recurring terminology, but it should not be confused with a complete semantic summary of the discussion.

5.8 Output and Integration Layer

The output layer makes the results available for subsequent analysis. Depending on the current implementation, the outputs may be stored in CSV files, held in data structures, or used directly by the dashboard. The integration stage should preserve the association between each analysis result and its source record or topic. Otherwise, sentiment or keyword results may be attributed to the wrong record. The dashboard can then present sentiment distributions, keyword summaries, or related analytical views where supported by the current application.

5.9 Workflow Diagram



Figure 6.2: NLP Processing Workflow

5.10 Architectural Considerations

The NLP component should be modular so that text preparation, sentiment analysis, keyword analysis, and output generation can be tested independently.

Modularity also simplifies future changes. For example, a transformer-based classifier could be evaluated as an alternative sentiment engine without redesigning the entire dashboard.

However, replacing VADER would require new dependencies, evaluation data, and performance testing.

CHAPTER 6
SOFTWARE AND HARDWARE SPECIFICATIONS

6.1 Dataset Overview

The dataset used by a social media analyzer generally contains records representing posts, comments, topics, or other units of online activity. The documented TrendPulse data structure includes text and engagement-related fields. Some project files are maintained as CSV datasets, including sample data and analyzed-data files. The final report should state the actual dataset source, collection method, time range, record count, and licensing conditions only after these details have been verified. A sample dataset must be identified as sample data rather than represented as a complete live feed.

6.2 Description of Important Fields

Platform: Identifies the social media platform associated with a record.

Topic: Represents the subject or category associated with the record.

Text: Contains the natural-language content used for sentiment and keyword analysis.

Likes: Represents the recorded number of likes.

Comments: Represents a comment-related engagement count, depending on the dataset definition.

Shares: Represents the recorded number of shares.

Views: Represents the recorded number of views.

The meaning of each field should be documented consistently. For example, a comments field may count comments on a post rather than the number of replies written by a particular user.

6.3 Importance of Data Quality

Poor data quality can affect NLP results. Missing text prevents meaningful sentiment analysis. Duplicated records can cause certain opinions or keywords to be counted repeatedly. Inconsistent encoding can damage punctuation or special characters. Incorrect topic assignments can produce misleading summaries. Data validation should therefore occur before results are interpreted.

6.4 Handling Missing Text

A missing text value should not be interpreted as neutral sentiment. Neutral sentiment means that the analyzer has processed text and assigned a neutral result according to its scoring rules. Missing text means that there was no usable textual input. These cases should be represented separately wherever the application supports such handling. An empty string should also be distinguished from a valid sentence that expresses a genuinely neutral statement.

6.5 Text Normalization

Text normalization makes text more consistent for analysis. Common operations include:

- Removing unnecessary leading and trailing whitespace.
- Normalizing repeated spaces.
- Handling missing values.
- Correcting encoding problems when possible.
- Applying consistent rules for duplicate detection.

Lowercasing and punctuation removal may help some keyword-analysis methods. However, these transformations can remove features that VADER uses to interpret sentiment intensity.

The original text should be preserved whenever possible so that preprocessing decisions can be reviewed.

6.6 Tokenization and Keyword Preparation

For frequency-based keyword analysis, text may be divided into tokens and filtered according to a defined policy. A typical workflow may include:

1. Selecting a text record.
2. Identifying word-like tokens.
3. Applying a consistent case policy.
4. Filtering irrelevant terms.
5. Counting term occurrences.
6. Aggregating counts across records.
7. Ranking or displaying selected terms.

This is a general methodology. The report should not claim that every step is implemented unless it is confirmed by the current project code.

6.7 Duplicate Records

Duplicate records can affect both sentiment proportions and keyword counts. If the same text is included repeatedly, it may receive disproportionate influence in an aggregate analysis.

Duplicate handling requires care because two legitimate users may independently post the same short expression. Exact-text matching is not always sufficient to determine whether two records represent the same underlying post. Where identifiers and timestamps are available, they may support better duplicate detection.

6.8 Preprocessing for Sentiment Versus Keywords

The preprocessing configuration should reflect the analytical objective. For sentiment analysis, negation, punctuation, capitalization, and emoticons may influence the score. Excessive cleaning can remove valuable signals. For keyword analysis, normalization and stop-word filtering may make recurring

content terms easier to identify. A useful design is to preserve the original text and create task-specific representations for sentiment and keyword processing.

6.9 Data Storage

CSV files are convenient for small and medium-sized academic projects because they can be inspected using spreadsheet software and processed using Python libraries. However, CSV files do not provide database transactions, relational constraints, or automatic concurrency management. For a larger deployment, a database or scalable data-processing platform may be more suitable. Such a change would require additional design and implementation work.

6.10 Dataset Documentation Requirements

Before final submission, the team should verify and document:

- Actual dataset source.
- Whether the data are live, collected, or illustrative.
- Total number of records.
- Date range.
- Text language or languages.
- Number of missing and duplicate records.
- Available metadata fields.
- Applicable data-use restrictions.
- Whether personally identifying information has been removed.

These details must be based on the actual dataset and should not be estimated or invented.

CHAPTER 7

PERFORMANCE EVALUATION PARAMETERS AND RESULTS

7.1 Introduction

Keyword analysis is the process of identifying words or phrases that help describe the content of a text collection.

In social media analytics, keywords can help users discover recurring subjects and examine the vocabulary associated with particular topics.

Within TrendPulse, keyword analysis complements sentiment analysis. Sentiment estimates the emotional orientation of text, while keywords help describe what the text discusses.

7.2 Purpose of Keyword Analysis

The main purposes include:

- Identifying frequently occurring terms.
- Summarizing the vocabulary associated with a topic.
- Supporting exploratory analysis.
- Highlighting terms that may warrant further investigation.
- Providing information for keyword-related dashboard visualizations.

A keyword is not necessarily a trend. A term may occur frequently because it is generic, duplicated, or part of a fixed phrase. Additional evidence is needed to establish whether interest in a term is increasing over time.

7.3 Term Frequency

Term Frequency (TF) measures how often a term occurs in a document or collection of documents. A simple collection-level term frequency can be written as: TF(t) = Number of occurrences of term t in the collection. For example, consider these three sentences:

1. “AI tools are useful.”
2. “AI tools can save time.”
3. “Many AI tools are available.”

The term “AI” appears in all three sentences. Its collection-level frequency is therefore three, assuming simple token counting and no additional filtering. This is an illustrative example, not a measured result from the TrendPulse dataset.

7.4 Document Frequency

Document Frequency (DF) measures the number of documents in which a term appears at least once. If "AI" appears in all three example sentences, its document frequency is three. Term frequency and document frequency are different. A term may appear many times in one document but occur in only one document overall. The distinction matters when the objective is to identify terms that are broadly distributed across a collection.

7.5 Stop-Word Filtering

Common words can dominate keyword lists without providing useful information.

For example, terms such as "the," "is," and "and" are often filtered in English keyword extraction. A stop-word list should be selected carefully. Words that appear common in one domain may be important in another. Negation words should not be removed from text used for sentiment analysis merely because they are common.

7.6 N-Grams

An n-gram is a sequence of n consecutive tokens.

Examples include:

- Unigram: "artificial"
- Bigram: "artificial intelligence"
- Trigram: "generative artificial intelligence"

N-grams can preserve phrases that lose meaning when split into individual words.

For example, the phrase "machine learning" may be more informative than counting "machine" and "learning" separately.

If phrase extraction is not part of the current implementation, n-gram analysis should be presented as a possible extension.

7.7 TF-IDF

Term Frequency–Inverse Document Frequency is a statistical method that estimates how important a term is to a document relative to a collection of documents. A common form is:

TF-IDF(t, d) = TF(t, d) × IDF(t) where TF(t, d) represents the frequency-related weight of term t in document d, and IDF(t) reduces the weight of terms that occur in many documents. A common inverse document frequency formulation is:

IDF(t) = log(N / DF(t))

where N is the total number of documents and DF(t) is the number of documents containing the term. Implementations may use smoothing and normalization. TF-IDF can help identify terms that are distinctive within a document or topic.

However, it should not be claimed as the current TrendPulse keyword algorithm unless the source code confirms that it is used.

7.8 Keyword Extraction Workflow

A general keyword-analysis workflow consists of the following steps:

1. Select the text collection.
2. Validate text values.
3. Normalize text according to the chosen method.
4. Tokenize the text.
5. Apply appropriate filtering.
6. Count or score terms.
7. Aggregate terms by topic or dataset where appropriate.
8. Rank or select terms.
9. Save or display the results.

The exact implementation may use a simpler or more advanced variation of this workflow.

7.9 Example

Consider the following illustrative comments:

- “AI tools help students learn.”
- “AI tools save time.”
- “Students use AI tools for coding.”

After suitable tokenization and stop-word filtering, a possible frequency table might look like this:

Term	Illustrative frequency
AI	3
tools	3
students	2
learn	1
time	1
coding	1

These counts assume a simple token-counting method and a particular normalization policy. They are not extracted from the actual TrendPulse dataset.

The table demonstrates how repeated terminology can be identified. It does not establish that “AI” is trending upward because no time-based comparison has been performed.

7.10 Keyword Analysis Versus Trend Detection

Keyword analysis identifies important terms within a dataset or time period. Trend detection examines how interest or activity changes over time.

For example, a term may appear 100 times in one dataset. This alone does not show whether its popularity is rising.

A more complete trend analysis might compare term frequency across successive time windows, account for changes in total record volume, and examine associated engagement metrics.

TrendPulse should therefore distinguish between recurring keywords and measured growth in attention.

7.11 Sentiment and Keywords Together

Keyword and sentiment information can be analyzed together.

For example, comments associated with a topic may frequently contain the term “update.” Sentiment analysis may then reveal whether the surrounding discussion is generally positive, negative, or neutral.

However, simply associating a keyword with a sentiment label does not prove that the keyword itself is positive or negative.

A stronger analysis would examine the text in which the term appears and consider the distribution of sentiment across relevant records.

7.12 Limitations of Keyword Analysis

Frequency-based keyword methods can be affected by duplicated text, spam, generic words, spelling variations, and context-dependent meaning.

They may also fail to recognize synonyms. For example, “AI,” “artificial intelligence,” and “machine intelligence” may refer to related concepts but appear as separate terms.

More advanced methods, such as phrase extraction, entity recognition, topic modelling, and embedding-based semantic similarity, could help address some of these limitations.

These extensions require separate implementation and evaluation.

CHAPTER 8

IMPLEMENTATION PLAN FOR NEXT SEMESTER AND FUTURE SCOPE

8.1 Future Scope

The NLP component can be improved through several extensions.

8.1.1 Transformer-Based Sentiment Analysis

Transformer-based models can produce contextual representations of text and may improve performance on complex sentences. A model such as a suitable pretrained BERT-family classifier could be evaluated against VADER on a labelled dataset. The decision to adopt such a model should consider computational cost, dependency requirements, inference speed, and measured performance.

8.1.2 Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis attempts to identify opinions about individual features of an entity. For example, a review may praise a product's design while criticizing its battery life. A single overall sentiment label may hide this distinction. Aspect-based analysis could provide more detailed information, but it requires additional methods and evaluation.

8.1.3 Multilingual Support

Future versions could support multiple languages through language identification and appropriate language-specific models. The system would need to be tested on representative examples for each supported language.

8.1.4 Advanced Keyword Extraction

The keyword component could be extended using TF-IDF, n-grams, named entity recognition, keyphrase extraction, or embedding-based similarity. These methods could help distinguish meaningful terms and phrases from generic vocabulary. The selected approach should be evaluated against the existing keyword method before replacement.

8.1.5 Topic Modelling

Topic modelling could help identify groups of related words and documents across a large text collection. Such methods may help reveal themes that are not captured by simple keyword frequency.

Topic modelling should be evaluated carefully because automatically generated topics may be difficult to interpret or may change with dataset composition.

8.1.6 Time-Based Trend Analysis

A future version could examine how keyword frequency, sentiment distribution, and engagement change over time. For example, a system could compare the number of relevant posts across successive time windows and normalize counts by total data volume. This would provide stronger evidence of increasing or decreasing attention than a single frequency count.

8.1.7 Improved Data Quality

Additional improvements could include better duplicate detection, language identification, spam filtering, and validation of incoming records. These measures could improve the reliability of downstream sentiment and keyword analysis.

8.1.8 Model Evaluation Dashboard

A future dashboard could show evaluation metrics, confusion matrices, processing time, and examples of classification errors. This would help developers assess model behavior and communicate its limitations more transparently.

8.2 Conclusion

Natural Language Processing provides useful techniques for converting unstructured social media text into information that can support data analysis.

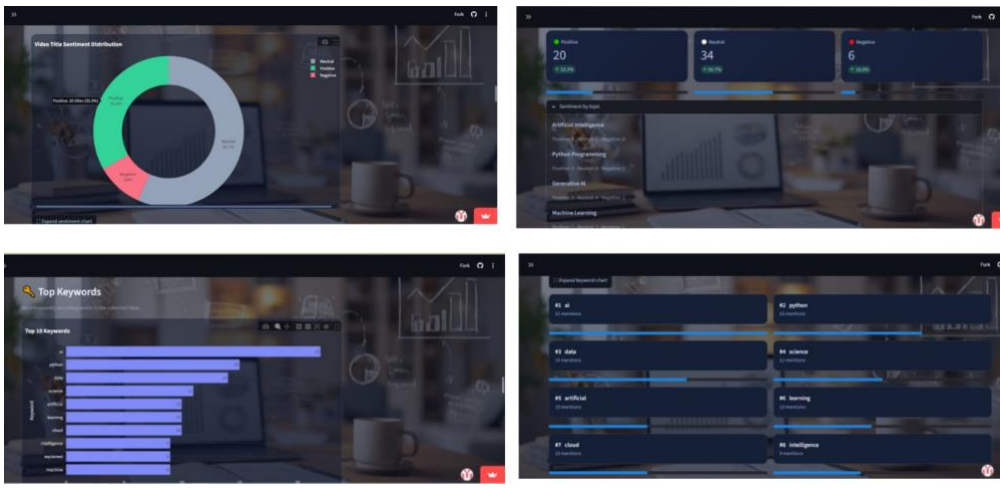
In TrendPulse, the documented NLP functionality focuses on sentiment analysis using VADER and keyword-oriented analysis. Sentiment analysis estimates the emotional orientation of text, while keyword analysis identifies recurring terms that may help summarize online discussions.

When integrated with topic and engagement information, these outputs can provide a broader view of social media activity than numerical engagement metrics alone.

The approach offers practical advantages for an academic project because it demonstrates the application of NLP concepts through accessible Python-based tools. It also provides a foundation for further investigation into contextual language models, multilingual processing, aspect-based sentiment analysis, and time-aware trend detection.

However, automated NLP results must be interpreted carefully. Lexicon-based sentiment analysis can misinterpret sarcasm and context, while keyword frequency alone cannot establish that a topic is becoming more popular. Reliable conclusions require appropriate data preparation, testing, and transparent reporting.

Overall, the NLP component demonstrates how text analysis can contribute to social media analytics while leaving opportunities for more advanced methods and rigorous evaluation in future development.



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